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Bowling Release Visual Analysis: Tracking and Reconstruction

Final Project
099993 - Image Analysis and Computer Vision

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1. Introduction

In modern sports analytics, precise kinematic metrics such as ball speed, entry angle, trajectory curvature, and spin rate are fundamental for evaluating player performance. In the sport of bowling, extracting these metrics is particularly critical, as the physical interaction between the ball's rotational dynamics and the localized oil pattern of the lane dictates the ultimate success of a shot [9].
Despite the importance of this data, extracting intricate parameters from live gameplay presents a significant technical barrier. Currently, commercial systems capable of extracting these metrics typically rely on complex, synchronized multi-camera stereoscopic setups or specialized sensor arrays [12]. While highly accurate, these proprietary systems are cost-prohibitive, restricting access strictly to professional broadcasting environments or high-end training facilities. Consequently, there is a growing demand to democratize sports analytics by developing robust computer vision systems capable of extracting deep metric insights from standard, accessible camera hardware.
However, transitioning from specialized sensor arrays to a single, uncalibrated static camera introduces a host of geometric and visual challenges. Because intrinsic camera parameters are typically unavailable in broadcast or consumer footage, resolving depth requires strict adherence to projective geometry and physical constraints. Furthermore, tracking fast-moving sports balls from a monocular feed presents multiple challenges due to motion blur, occlusion, and background clutter, making the calculation of sub-pixel rotational velocity (spin) highly complex.
To bridge this gap, this report proposes a comprehensive, monocular computer vision pipeline designed to extract robust 3D trajectory and spin parameters from standard bowling video footage.

2. Problem Formulation

2.1. Problem Statement

The primary objective of this project is to estimate the 3D trajectory and spin dynamics of a bowling ball from a static monocular video feed for advanced sports analysis. Because intrinsic camera parameters are not provided, the system must rely on projective geometry and physical constraints.
To achieve a robust and accurate estimation, the problem is formulated as a structured computer vision pipeline consisting of five interdependent tasks:

1. **Lane Detection:** Identifying the physical boundaries of the playing surface within the image plane, despite perspective distortion and object occlusion.
2. **Ball Trajectory Tracking:** Isolating the bowling ball across consecutive frames and applying temporal interpolation to generate a continuous, robust 2D trajectory from discrete, raw center coordinates.
3. **Ball Spin Estimation:** Analyzing the rotational displacement of surface features on the ball to estimate its spin rate, a critical parameter in bowling performance.
4. **Planar Metric Rectification:** Utilizing the detected lane boundaries to compute a planar homography matrix, mapping the perspective-distorted image space to a metric, 2D bird's-eye view.
5. **3D Trajectory Reconstruction:** Lifting the 2D tracking data into 3D spatial coordinates by enforcing two geometric constraints: the ball maintains constant contact with the rectified lane plane, and it possesses a known, fixed physical radius.

2.2. State of the Art

2.2.1 Lane Detection and Metric Rectification

The foundational techniques for boundary extraction are heavily derived from Advanced Driver Assistance Systems (ADAS). Standard pipelines in dynamic environments rely on localized gradient filters (such as the Sobel or Canny operators) and the Hough Transform to extract linear boundaries [1].
However, translating pixel-space detections into metric measurements traditionally requires full camera calibration [30]. In structured sports environments, where the playing surface is guaranteed to be a flat plane and the camera is static, the state of the art often bypasses explicit calibration in favor of Planar Homography [13]. By identifying four coplanar control points with known real-world dimensions (such as the corners of a tennis court or bowling lane), a homography matrix can be computed to rectify the image, providing a highly accurate metric mapping of the surface [29].

2.2.2 Ball Tracking, Trajectory Estimation, and Spin

Tracking fast-moving sports balls is challenging due to motion blur, occlusion, and background clutter. Modern solutions frequently utilize deep learning frameworks (e.g., TrackNet[14]) or background subtraction to isolate the ball. To transition from 2D pixel tracking to 3D spatial estimation without depth sensors, domain-specific applications enforce physical priors. By leveraging the geometric constraint that a rigid sphere maintains contact with a known ground plane, the object's spatial coordinates can be robustly estimated [5]. Furthermore, estimating spin from a monocular feed is notoriously difficult. State-of-the-art approaches typically involve analyzing the frame-by-frame displacement of surface textures (such as finger holes, logos, or dotted patterns) using feature-matching algorithms, template matching, or optical flow [11] [18]. Recent studies have even explored event-based cameras to capture ultra-fast spin dynamics [21], though adapting texture-based tracking for standard frame-rate cameras remains the primary method for accessible sports analytics.

2.3. Solution Approach

The first phase of the pipeline focuses on resolving the spatial geometry of the scene. Due to severe occlusion and the nature of a bowling alley, the playing surface is localized by independently extracting its distinct geometric boundaries (the foul line, the lateral gutters, and the pin deck) using a combination of directional gradient analysis, spatial proximity heuristics, and robust object-centric regression. The intersection of these boundaries yields four coplanar control points in the image plane.
Following lane boundary extraction, the second phase isolates the kinematic trajectory of the bowling ball. The ball is tracked across the temporal sequence to extract its raw 2D center coordinates and apparent pixel radius. To mitigate the fragmentation of data caused by fast motion and object occlusion, temporal interpolation is applied to generate a continuous, smooth mathematical curve in the image plane.
To provide comprehensive sports analytics, the system evaluates the rotational dynamics of the ball. By tracking the cyclical, frame-by-frame spatial displacement of high-contrast surface textures across the ball's visible hemisphere, the angular velocity is mathematically estimated, yielding the overall spin rate over the course of the trajectory.
Then, by mapping the homogeneous pixel coordinates of the lane boundaries (in the image plane) to the known metric dimensions of a standard bowling lane, a 2D-to-2D planar homography matrix (H) is computed. This mathematical transformation effectively eliminates perspective distortion, providing a rectified, metric bird's-eye view of the playing surface without requiring intrinsic camera calibration.
Finally, to reconstruct the 3D trajectory, the ball's image-plane contact point (found by adding its pixel radius to its vertical center) is projected through the homography matrix to obtain its metric ground coordinates (X,

Y). By enforcing the constraint that the ball maintains surface contact with the lane, its known physical radius is simply assigned as the Z-coordinate, yielding the final spatial position (X, Y, R).

3. Methodology

3.1. Lane Detection

Lane detection constitutes a fundamental component of the proposed bowling analysis pipeline. Unlike traditional autonomous driving applications, where lane detection is primarily used for navigation, in this work it serves two main purposes: restricting the search region for bowling ball detection and estimating the lane geometry required for perspective rectification. By estimating the lane boundaries, a planar homography can be computed to rectify the image plane into a top-down representation of the bowling lane. This rectification, combined with the assumption that the bowling ball remains in contact with the lane surface and preserves a constant radius, enables the estimation of the ball trajectory in three-dimensional space.
Bowling lane detection presents several challenges, including varying illumination conditions, strong perspective effects, the presence of multiple adjacent lanes with similar geometric structures, and partial occlusions caused by bowling pins. Furthermore, not all lane boundaries exhibit the same visual characteristics. While some borders appear as strong visible edges, others are partially occluded or difficult to identify directly. For this reason, a boundary-specific detection strategy was adopted, where each lane border is estimated using a tailored approach while maintaining a shared preprocessing pipeline.
Since the camera remains fixed throughout each bowling recording, the lane geometry is assumed to remain constant across frames. Consequently, lane boundaries are estimated only once using the first frame of the video, and the resulting geometric configuration is reused for the subsequent rectification and trajectory reconstruction stages. This assumption substantially reduces computational cost, since the camera remains fixed throughout the recording.
The proposed lane detection framework consists of three main stages: image preprocessing, edge extraction, and geometric line estimation. Standard preprocessing operations, including color space transformation and Gaussian smoothing, are first applied to enhance relevant structural information and suppress noise. Subsequently, edge detection and Hough-based line extraction are employed, with specialized heuristics introduced depending on the boundary being estimated.

3.1.1 Single-Channel Image Transformation

Since edge detection methods operate on single-channel intensity images, an initial transformation from the RGB image space $R^{n\times m\times 3}$ to a single-channel representation $R^{n\times m}$ is required. The effectiveness of this transformation directly affects the separability of lane boundaries from the surrounding scene, influencing both edge detection robustness and the sensitivity of subsequent line extraction methods. The objective of this stage is to maximize the contrast between relevant lane structures and the background while minimizing the effect of illumination variability across videos. Bowling recordings may differ considerably in lighting conditions, reflections on the lane surface, and camera exposure settings, making the choice of transformation particularly important.
To identify the optimal preprocessing step, several distinct single-channel conversion methods were evaluated. As a baseline, the standard OpenCV grayscale conversion was tested, which computes a weighted combination of the RGB channels according to perceptual luminance [10, 22]

$$Y = 0.299\,R + 0.587\,G + 0.114\,B \tag{1}$$

where R, G, and B represent the red, green, and blue image channels, respectively.
An alternative standard structural representation, lightness [22], was also tested to observe if averaging the maximum and minimum color components provided better separability for the lane textures. The lightness channel from the HLS color space is mathematically defined as:

$$L = \frac{\max(R, G, B) + \min(R, G, B)}{2}$$

Beyond standard conversions, domain-specific approaches were explored. A color transformation inspired by lane detection approaches in the autonomous driving domain [28], calculated as

$$I = R + G - B \tag{2}$$

was evaluated. This transformation emphasizes red and green channel intensities while suppressing the blue channel, theoretically increasing the contrast between bright lane markings and darker road-like surfaces.

To further improve robustness against illumination changes, Principal Component Analysis (PCA) [15] was integrated into the preprocessing pipeline. The rationale for this approach was drawn from the field of Earth Observation and remote sensing, where PCA is widely utilized to transform highly correlated multispectral satellite bands into a concise subset of structural predictors [24]. By treating the standard RGB video feed as a three-band multispectral image, the color channels were projected into a lower-dimensional space. Retaining only the first principal component mathematically maximizes the statistical variance of the scene. This data-driven projection effectively isolates the dominant physical structures of the environment while naturally suppressing correlated noise like shadows, surface reflections, and ambient lighting artifacts.
Finally, to assess whether boundary transitions could be isolated strictly through chromaticity rather than intensity, the hue channel from the HSV color space was evaluated.
To determine the most robust approach for the final pipeline, the five single-channel transformations were systematically evaluated against various edge detectors (Sobel[26], Canny[4], and Laplacian[19]) to measure their impact on boundary localization. Preliminary evaluations indicated that PCA, when paired with a Sobel filter, provided the most robust and consistent boundary localization across the varying illumination conditions of the dataset. The detailed quantitative analysis and discussion of this evaluation are presented in section 4.1. Consequently, the PCA transformation paired with Sobel edge detection was selected as the foundational preprocessing step for the lane detection architecture.

3.1.2 Image Smoothing

Before edge detection, Gaussian blurring is applied to reduce high-frequency noise and suppress small intensity variations that may generate false edge responses. In bowling lane imagery, reflections on the lane surface and texture variations can introduce spurious gradients that negatively affect boundary detection. A Gaussian filter with kernel size 5×5 is applied to the single-channel transformed image. This preprocessing step improves the stability of subsequent edge detection methods while preserving the dominant structural information of the lane boundaries.

3.1.3 Edge Detection

After image smoothing, edge detection is applied to extract candidate structures corresponding to lane boundaries. Three classical gradient-based and intensity-based methods were evaluated: Sobel, Canny, and Laplacian filters. These methods differ in how they estimate intensity discontinuities and in their sensitivity to noise and texture variations.
The Sobel operator computes first-order image gradients and is used to approximate horizontal and vertical intensity changes. It is particularly useful for emphasizing directional structures, making it suitable for detecting lane boundaries with a strong geometric orientation.
The Canny detector follows a multi-stage process involving gradient computation, non-maximum suppression, and hysteresis thresholding. This approach typically produces thinner and more connected edges, but its performance is sensitive to threshold selection and illumination variability.
The Laplacian operator is a second-order derivative method that detects regions of rapid intensity change regardless of direction. While it can highlight fine structures, it is also more sensitive to noise and may produce fragmented edge maps in low-contrast regions.
All three methods were implemented and evaluated across multiple bowling videos. Based on qualitative performance and stability of the resulting edge maps, the Sobel operator was selected as the primary edge detector for the proposed pipeline. Depending on the boundary being estimated, the gradient direction is adapted to emphasize either horizontal or vertical structures, improving the extraction of relevant lane edges prior to line fitting.

3.1.4 Line Extraction Using the Hough Transform

Following edge detection, candidate line segments corresponding to lane boundaries are extracted using the probabilistic Hough transform [20]. The Hough transform [7] is particularly suitable for lane detection tasks, as it enables robust identification of line structures even when edges are fragmented or partially incomplete. The method operates by mapping edge points from the image space into a parametric representation of lines. Since infinitely many lines may pass through a given image point, each edge pixel votes for all compatible line hypotheses in the parameter space. Line candidates receiving strong consensus from multiple edge points accumulate higher votes and are identified as probable geometric structures. In this work, the probabilistic Hough transform (HoughLinesP) is employed. Unlike the standard Hough transform, which returns infinite lines represented in polar form (ρ, θ), the probabilistic formulation directly estimates finite line segments defined by their endpoints (x_1, y_1, x_2, y_2). This representation is particularly advantageous for the proposed pipeline, as subsequent processing relies on geometric properties such as segment orientation, intersections, and proximity to expected lane locations. Additionally, the probabilistic formulation reduces computational cost by operating on a subset of edge points while maintaining robust detection performance. The resulting line candidates are further filtered according to geometric constraints consistent with the expected lane structure. In particular, line orientation is used to reject candidates that deviate substantially from the expected direction of the corresponding lane boundary. Since multiple plausible line hypotheses may still remain, additional boundary-specific heuristics are introduced in later stages of the pipeline to determine the final lane borders.

3.1.5 Bottom Boundary Detection

The bottom boundary of the bowling lane corresponds to the nearest and most visually distinctive lane edge in the scene. Unlike other lane borders, this boundary is typically less affected by occlusion and exhibits a strong horizontal structure, making it the most reliable reference for initial lane geometry estimation. To improve robustness and reduce interference from irrelevant scene content, a spatial cropping is first applied to focus on the lower-central region of the frame, where the bottom boundary is expected to appear. This restriction reduces noise from distant lane regions and peripheral elements. The cropped region is then converted into a single-channel representation and smoothed using Gaussian filtering. Edge detection is performed using a Sobel operator configured to emphasize horizontal intensity changes, which are most consistent with the orientation of the bottom lane boundary. After edge extraction, the probabilistic Hough transform is applied to identify candidate line segments. Since multiple line hypotheses may be generated due to reflections and lane texture, the resulting segments are filtered based on geometric constraints. In particular, only lines with slopes within a predefined threshold are retained, ensuring consistency with the expected near-horizontal orientation of the bottom boundary. Among the remaining candidates, the most stable line is selected as the bottom boundary estimate.

3.1.6 Lateral Boundary Detection

The lateral boundaries of the bowling lane present a significantly more challenging detection problem compared to the bottom boundary. Unlike the single dominant horizontal edge observed at the bottom of the lane, the lateral region contains multiple parallel lane markings across adjacent lanes. This introduces ambiguity, as several valid line candidates may be detected in the image, and the correct pair of boundaries depends on the specific lane being analyzed. To address this, the same preprocessing pipeline is applied, consisting of single-channel transformation, Gaussian smoothing, and edge detection using a Sobel operator configured to emphasize vertical structures. However, to mitigate noise generated by architectural clutter on the background wall at the far end of the lane, a spatial crop is applied, removing the top 15% and bottom 10% of the frame prior to edge extraction.

After edge extraction, the probabilistic Hough transform is used to obtain candidate line segments. A preliminary geometric filter is applied to discard near-horizontal segments, ensuring that only approximately vertical structures are retained. Since multiple valid vertical line hypotheses typically remain, an additional selection strategy is required. To disambiguate between adjacent parallel lines and isolate the target lane, the pipeline is designed to accept an intra-lane reference coordinate as an initialization parameter. Rather than hardcoding a fragile spatial assumption (such as the exact image center), treating this coordinate as a configurable input allows the algorithm to adapt to various camera angles and lateral offsets. In the absence of an explicitly provided parameter, the script falls back to a default central point. However, initializing the algorithm with a known coordinate guarantees that the lane detection focuses on the correct lane. A horizontal line passing through this reference point is then considered, and intersections between this line and all detected line candidates are computed. Each candidate line is evaluated based on the horizontal distance between its intersection point and the reference point. The left and right lane boundaries are selected as the closest valid lines on each side of the reference point, respectively. This heuristic effectively resolves the ambiguity introduced by multiple parallel lane structures by converting the problem into a geometric proximity selection task in one dimension.

3.1.7 Top Boundary Detection

The top boundary of the bowling lane presents a fundamentally different challenge compared to the other lane borders, as it is frequently occluded by bowling pins. This occlusion breaks the continuity of the lane marking and makes classical edge-based or line-based methods such as the Hough transform unreliable. Additionally, feature-based approaches such as Scale Invariant Feature Transform (SIFT)[17] were considered but found to be ineffective due to the small size of the pins in the image and the limited number of stable keypoints they generate at the typical camera distance.

To address these limitations, a template matching-based approach is adopted, leveraging prior knowledge of the approximate appearance and scale of bowling pins. The method searches for occurrences of a pin template within the upper region of the frame, where pins are expected to appear. Since the apparent size of pins varies

across videos due to perspective and camera distance, a multi-scale search strategy is applied. The template is resized across a range of scales, and matching is performed at each scale independently. This increases robustness to variations in pin size and improves detection consistency across different recordings.
After matching, multiple overlapping detections are typically produced for the same physical pin. To remove redundant detections, non-maximum suppression is applied, retaining only the most confident bounding boxes. From each remaining detection, representative keypoints are extracted, corresponding either to the center or the lower portion of the bounding box, depending on the selected configuration.
Given that template matching can still produce false positives in visually similar regions, a robust geometric fitting stage is applied. The extracted keypoints are first filtered using Random Sample Consensus (RANSAC)[8] regression to remove outliers. A final first-order polynomial is then fitted to the inlier points using least squares, resulting in a continuous estimate of the top lane boundary.

3.1.8 Lane Boundary Integration and Corner Estimation

Once the bottom, lateral, and top lane boundaries are estimated independently, they are combined to recover a consistent geometric representation of the bowling lane. Since each boundary is represented as a line segment in image space, the lane corners can be obtained by computing the pairwise intersections between adjacent boundaries.
Specifically, the bottom and top boundaries are intersected with the left and right lateral boundaries to obtain the four lane corners: bottom-left, bottom-right, top-right, and top-left. These intersection points define a quadrilateral that approximates the perspective projection of the bowling lane in the image.

3.2. Ball Dynamics Estimation

The products derived from the bowling release are the ball's 2D trajectory along the lane and its spin. The trajectory is defined by localizing the geometric center of the ball through the clip's frames, whereas the parameters of spin are axis orientation and angular velocity. The previously obtained data is used to define the region of interest in which we'll find the ball's projection as a circle in every frame. Once the trajectory data is available, the ball's locations across consecutive frames are used to determine the positional change of its features and to derive its spin parameters. With both products in hand, we can then generate a visualization that consistently follows the ball's motion and proceed with the 3D reconstruction of the scene.

3.2.1 Trajectory Tracking

The ball-tracking workflow consists of three main steps: video preprocessing, ball-candidate detection, and trajectory computation. We begin by isolating the scene's region of interest and applying noise-reduction and contrast-enhancement filters to each frame. Then, the *Circle Hough Transform* (CHT) algorithm is applied to generate a dataset of all geometries that could represent the ball's position in each frame. Finally, we select the geometries that best form a coherent trajectory from the candidates, interpolate any missing values in the set, and smooth the final trajectory. In the end, we obtain a ball-center dataset with the image geometric coordinates and their corresponding radii.
Firstly, each frame must be manipulated to reduce video noise so that the CHT consistently detects the ball. This process starts by masking all pixels outside the ROI using the coordinates retrieved in section 3.1. Then, median blurring is applied to remove high-frequency noise from light reflections in the lane, followed by adaptive histogram equalization to increase contrast specifically in the region where the ball is located. This last step is particularly helpful because the ROI is expected to have a relatively consistent surface across the frames. We end up with a black-and-white clip in which the ball is clearly distinguishable from the rest of the scene, and the lane reflections are smoothed out.

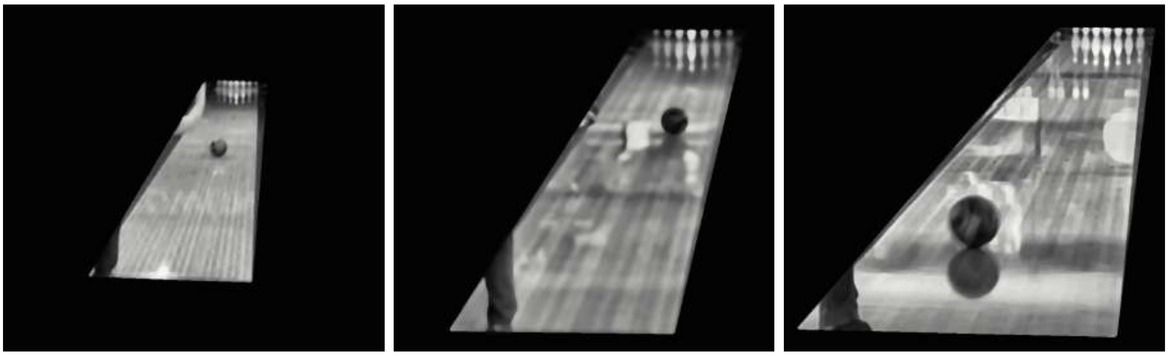

Figure 1: Preprocessed frames of different bowling releases. The ball is clearly distinguishable from the rest of the scene, and the lane imperfections are smoothed out.

Once we have the preprocessed clip, we apply OpenCV's built-in CHT algorithm. The library implements the *Hough gradient method*, which first applies *Canny edge detection* and then computes the local gradient for every nonzero pixel via the first-order Sobel x and y derivatives. These gradients are used to populate the two-dimensional *accumulator plane* [3], in which potential circle centers are tracked. For every edge pixel, a line is projected along its gradient from a specified minimum to a maximum distance, incrementing the accumulator cells along the way. The local maxima in this plane are selected as candidate centers. Finally, for each center, the algorithm evaluates the distances to nearby edge pixels to determine the optimal radius, preserving the circle only if it receives sufficient support and maintains a minimum distance from previously validated centers. The result is a dataset containing every detected circle, its x- and y-center coordinates, and its radius for each frame.

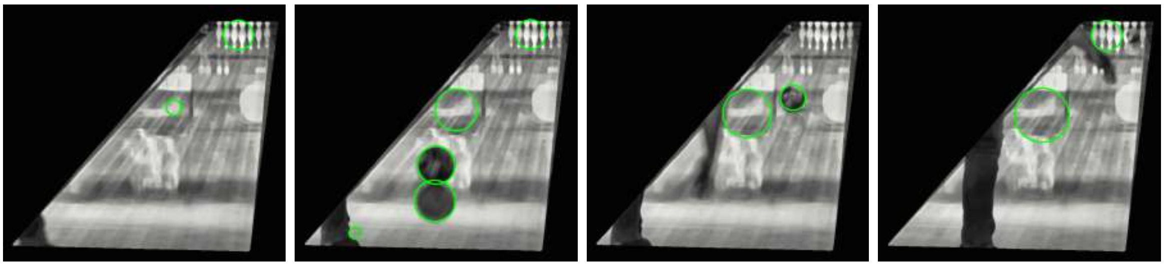

Figure 2: Detected circles overlayed on the preprocessed frames of the same clip. The algorithm detects multiple circles in each frame, which may include false positives due to noise.

This algorithm detects circles across video frames; however, it may also introduce noise due to numerical instabilities [3], leading to false detections, or even no detections over the ball, as seen in figure 2. The third step of the workflow focuses on filtering all circles representing the ball in each frame, resulting in a final defined trajectory. The methodology for discernment was inspired by [23], in which they elaborate a weighted directed graph to determine the ball's path in a soccer broadcast. The difference in our implementation is that a bowling release is less dynamic than a soccer kick; its behavior is more predictable, and its trajectory is easier to generalize.

The main idea is that each ball candidate in a frame is represented by a graph node, which is then directionally connected to all other candidates in the five subsequent frames. This is done because the previous step might miss the ball in consecutive frames. In the end, the longest path in the graph is retrieved, which represents the actual ball trajectory. To fill in the missing frames in the trajectory, cubic spline interpolation is applied to both the x- and y-center coordinates, followed by a light Gaussian filter ($\sigma = 5$) to eliminate noise, and a simple quadratic least squares interpolation for the radius values.

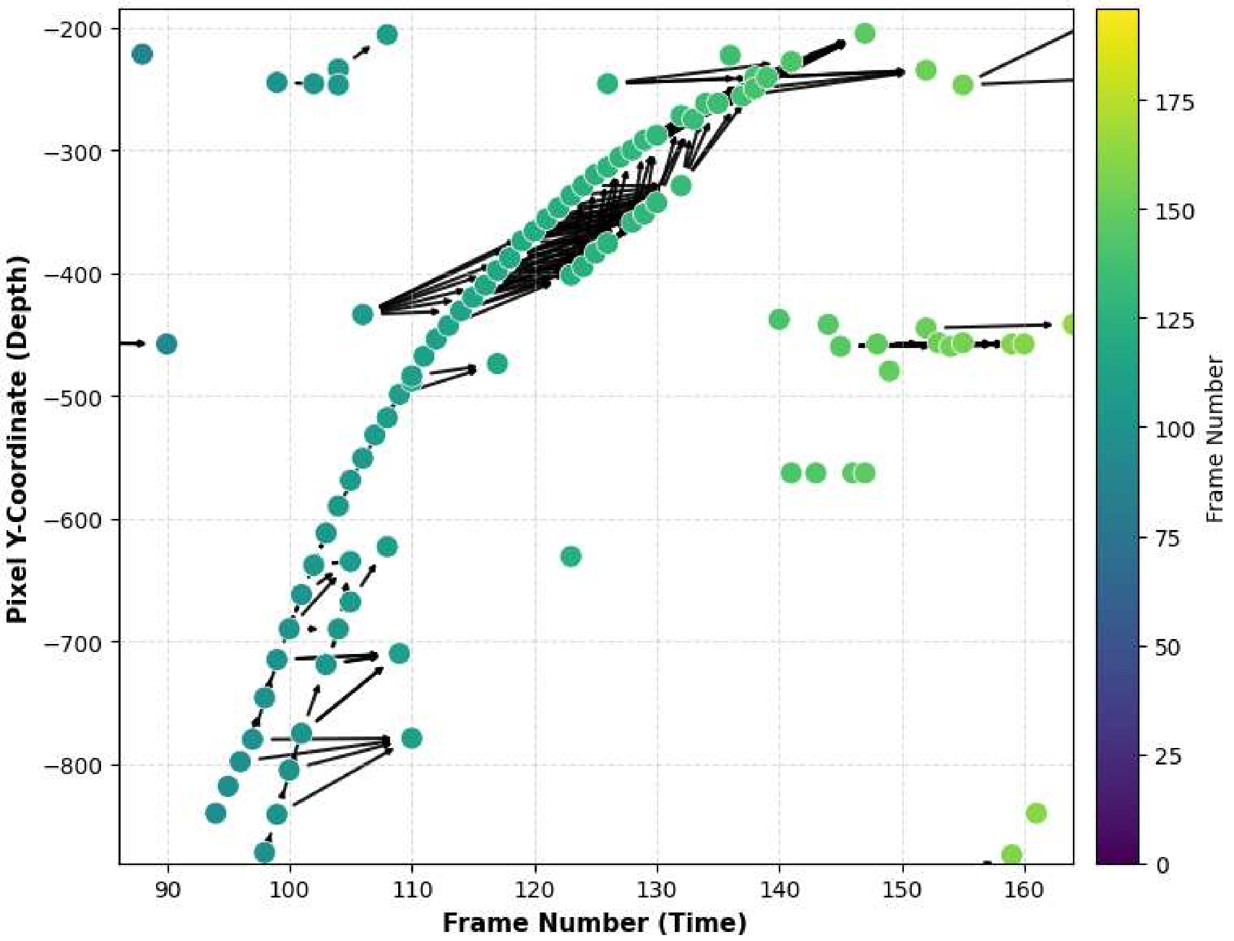

Figure 3: Graph representation of the ball candidates across frames. Each node represents a candidate circle, and directed edges connect candidates in consecutive frames. The longest path through the graph corresponds to the most coherent trajectory of the ball.

The entire workflow yields the final dataset, which contains the coordinates and radius of each circle from the first to the last detected frame of the clip. This data will be crucial for the following steps in spin detection and 3D reconstruction.

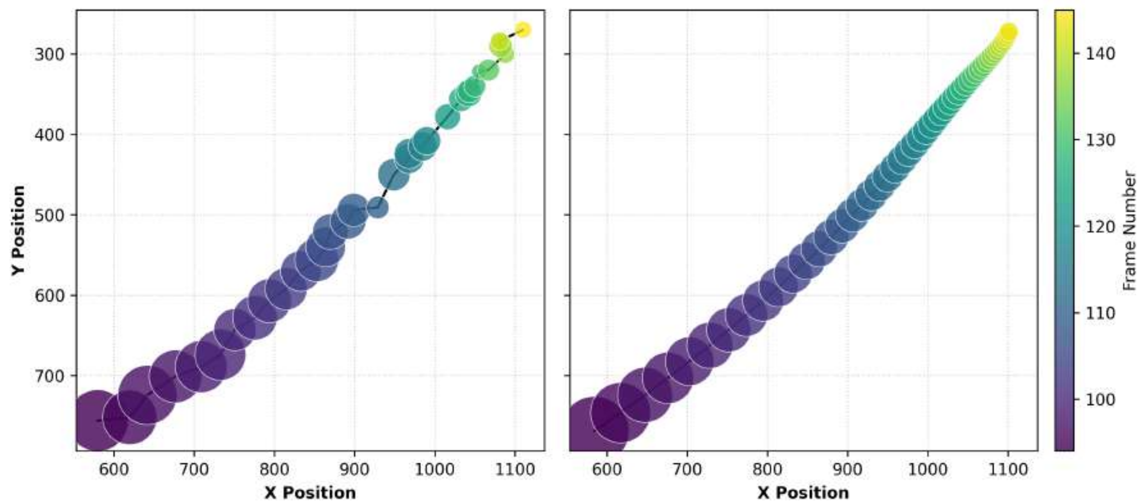

Figure 4: Left: Observed ball trajectory, with circle size proportional to observed radius. Right: Final trajectory and radii after interpolation and smoothing.

3.2.2 Spin Estimation

The motion describing the spin of the bowling ball is a rigid body rotation with respect to a reference frame fixed to its geometric center, fully characterized by a rotation axis and an angular velocity around it. The spin detection workflow consists of four main steps: bounding box definition, feature detection, optical flow estimation, and kinematic solving. First, the trajectory data obtained in the previous section is used to define a dynamic bounding box around the ball in each frame, which constrains the region of interest for the subsequent analysis. Then, trackable features are identified on the ball's surface using the Shi-Tomasi corner detection algorithm. Their displacement between consecutive frames is then estimated via the Pyramidal Lucas-Kanade optical flow algorithm, yielding a 2D tracked-features dataset. Finally, the 2D correspondences are projected onto the ball's hemisphere and fed into the Kabsch algorithm to recover the 3D rotation matrix, from which the spin axis and angular velocity are extracted. It is important to note that the spin detection workflow relies on detected features on the ball's surface, which are expected to be non-homogeneous. Therefore, the system is designed to work with a textured ball, as a homogeneous surface would not provide sufficient features for tracking.

The region of interest is defined as a bounding box around the detected ball. The optical flow requires two consecutive frames with the same dimensions, cropped to focus only on the ball. Therefore, robust trajectory tracking must be implemented prior to this step to avoid analyzing sections where the ball does not appear or is only partially visible. As the ball's radius decreases monotonically throughout the video, the bounding box dimensions are calculated dynamically, considering the ball's diameter of the first frame plus slight padding. Finally, grayscaling is applied to both frames.

Once we have defined the bounding box, we select the trackable features on the ball's surface. To compute optical flow, we identify points in the first image that are sufficiently distinct so they can be located again in the next image. OpenCV provides a pipeline based on the Shi-Tomasi corner detection algorithm [25]. Instead of relying solely on edge information, which captures gradient changes in a single direction, the algorithm seeks regions with significant intensity variation in two orthogonal directions. It computes the first-order image gradients, I_x and I_y, using a 3x3 Sobel kernel, which represent the rate of change of brightness in the horizontal and vertical directions. These gradients are then used to compute the *structure tensor* (or second-moment matrix) M over a window function w that captures the spatial context around a pixel:

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \tag{3}$$

The structure tensor is a covariance matrix of the image gradients. Its eigenvalues λ_1 and λ_2 are the intensity change in each direction. If both are small, the region is flat; if one is larger than the other, it represents an edge; and if both are large, we have found the so-called corner, where there is a strong intensity change in both orthogonal directions. Corners typically represent the best trackable features for the optical flow. In this context, the intensity changes must be detected solely on the ball's surface. Hence, the importance of the ball not having a homogeneous texture. We finally end up with an array of pixel locations that hopefully will be found in the next frame.

With the initial features detected, we approach their displacement between consecutive frames using the *Pyramidal Lucas-Kanade* (PyLK) optical flow algorithm [2]. The standard Lucas-Kanade algorithm rests on three core assumptions about the scene: brightness constancy, spatial coherence, and temporal persistence. The first implies that the brightness of tracked pixels does not change between frames. Spatial coherence means that neighboring points in a scene belong to the same surface and have similar motion. The last assumption is that features move relatively small distances between consecutive frames, as it relies only on a local window of information [3]. The latter, however, conflicts with the behavior of the spinning ball: the ball's rotational speed causes features to move noticeably between frames.

We want to determine the velocity vectors that indicate how far the feature has moved in the x and y directions. The brightness constancy tells us that the pixel intensity $I(x, y, t)$ at a specific point on the bowling ball does not change as the ball moves to a new position in the next frame:

$$I(x, y, t) = I(x + \Delta x, y + \Delta y, t + 1) \tag{4}$$

By linearizing, we obtain the *standard optical flow equation*:

$$I_x u + I_y v = -I_t \tag{5}$$

, where I_t is the difference in pixel intensity between frames, I_x and I_y are the spatial image gradients (computed in the feature detection), and u and v are the unknown velocity vectors. However, as we have a single equation for two unknowns, it cannot be solved with a single pixel. The solution takes advantage of the spatial coherence assumption: the optical flow is calculated for all the pixels around the feature for which we can build a system solvable by least-squares minimization:

$$\begin{bmatrix} I_x(p_1) & I_y(p_1) \\ I_x(p_2) & I_y(p_2) \\ \vdots & \vdots \\ I_x(p_n) & I_y(p_n) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} I_t(p_1) \\ I_t(p_2) \\ \vdots \\ I_t(p_n) \end{bmatrix} \tag{6}$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = (A^T A)^{-1} A^T b \tag{7}$$

The standard Lucas-Kanade algorithm is effective for small displacements but struggles with larger movements, as in our spinning ball. The local window in which the algorithm operates may not contain the corresponding feature in the next frame, causing it to be missed. Making the window larger, however, conflicts with the spatial coherence assumption, as the pixels might belong to different objects and move in multiple directions. The pyramidal implementation of the algorithm allows the handling of larger displacements by creating a multi-scale representation of the image. First, optical flow is computed at a lower resolution, and then the motion estimates are iteratively used to make the next guess at a higher resolution until the original scale is reached [2].

The Pyramidal Lucas-Kanade implementation provided by OpenCV is mathematically robust for optical flow estimation. However, the output may include false positives, as it always returns a destination coordinate for a feature, even if the feature becomes occluded, leaves the frame, or is distorted by motion blur. To filter out these falsely tracked features, we implement *forward-backward validation* [25]. We calculate the forward optical flow by projecting the corners from the first frame p_0 onto their estimated positions in the subsequent one p_1. For the backward optical flow, the new estimated positions are projected to new initial positions in the initial frame $p_0\prime$, which should match the original corners. We calculate the error between p_0 and $p_0\prime$, which should not exceed a predefined threshold to keep the tracked feature.

$$E_{FB} = |p_0 - p_0'| \tag{8}$$

This step is particularly important because there are ever-present non-rotating reflections on the ball's surface, which trigger multiple misses in the optical flow.

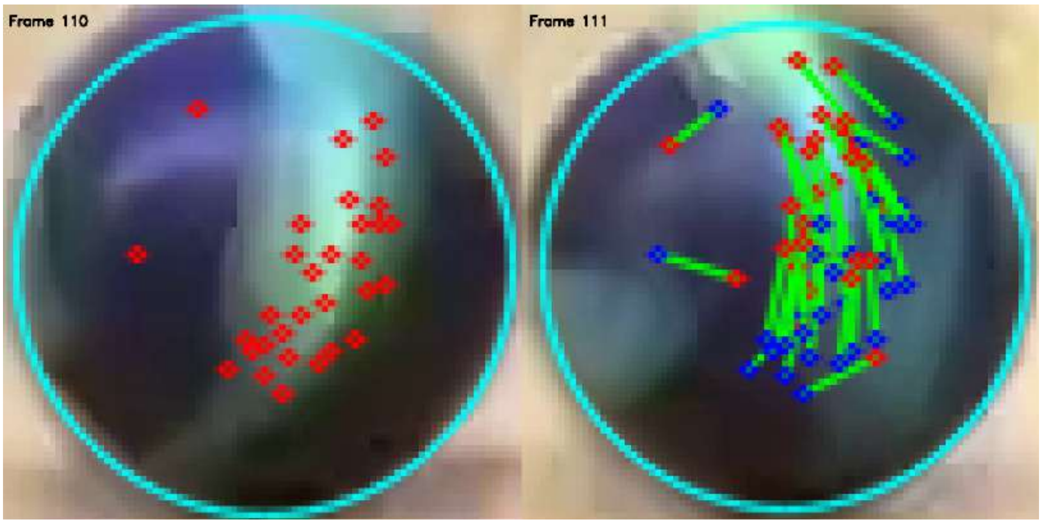

Figure 5: Subsequent frames optical flow. Left: First frame's features detected on the ball's surface. Right: Next frame's feature tracked positions.

The optical flow output yields a 2D tracked-features dataset, containing an initial point cloud P_{2D} and its corresponding final point cloud Q_{2D} for every pair of consecutive frames. Now we need to transform those 2D (u, v) coordinates into 3D space (X, Y, Z). The bounding box's center (C_x, C_y) coincides with the ball's geometric center. We subtract it from (u, v) to obtain 2D local X and Y coordinates in a ball-centric reference frame. Then, these coordinates can be projected onto the ball's hemisphere, assuming it is a perfect sphere and that $r \approx \frac{1}{2}\texttt{frame\_width}$ (defined in the bounding box). By solving the sphere equation we get the missing Z coordinate:

$$\begin{aligned} r^2 &= X^2 + Y^2 + Z^2 \\ Z &= \sqrt{r^2 - X^2 - Y^2} \end{aligned} \tag{9}$$

With the 3D point cloud P_{3D} and its corresponding displacement Q_{3D}, the next step is to compute the rotation that aligns P into Q. This is known as the *orthogonal Procrustes* problem, which we resolve by implementing the *Kabsch algorithm* [16]. First we compute the cross-covariance matrix H, which encapsulates the correlation between point clouds, and factor it using *singlular value decomposition* to get the orthogonal matrices U and V^T, that represent the rotations, and the diagonal matrix of singular values Σ:

$$\begin{aligned} H &= P^T Q \\ \mathrm{SVD}(H) &= U \Sigma V^T \end{aligned} \tag{10}$$

Having extracted the orthogonal components, the optimal rotation matrix R is computed as VU^T. However, because SVD is a purely mathematical factorization, it may yield a matrix that implies a coordinate inversion across the plane rather than physical rotation. This is identified by a negative determinant of R, so we invert the sign of the third column of V and recompute R whenever $\det(R) = -1$:

$$R = VU^T \tag{11}$$

While the matrix R fully describes the orientation change, it is not directly interpretable in terms of physical kinematics. Therefore, we convert R into an axis-angle representation. The angular velocity (the magnitude of rotation per frame), θ, is derived from the trace of the rotation matrix:

$$\theta = \arccos\left(\frac{\mathrm{tr}(R) - 1}{2}\right) \tag{12}$$

Finally, the 3D unit vector $[X, Y, Z]$ defining the stable rotational core of the ball (the spin axis) is extracted from the off-diagonal elements of R:

$$\mathrm{Axis} = \frac{1}{2\sin(\theta)} \begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix} \tag{13}$$

The kinematic solver outputs a frame-by-frame estimation of the raw 3D spin axis and angular velocity. However, this raw data is inherently corrupted by high-frequency noise from the optical flow process, as the number and direction of features are not always consistent and can yield inaccurate results. The final postprocessing workflow addresses axis-sign ambiguity correction, outlier removal, interpolation, and smoothing.

3.3. Lane Rectification and 3D Reconstruction

Once both lane geometry and ball trajectory have been estimated in the image plane, the next step consists of transforming the scene into a physically meaningful coordinate system aligned with the real-world dimensions of the bowling lane. Due to perspective distortion, measurements in the image space are not directly interpretable in metric terms. For this reason, a perspective rectification is performed using a homography grounded in the known geometry of the lane.

3.3.1 Planar Metric Rectification

Given the four lane corner points obtained from the intersection of the estimated boundaries, a planar homography is computed to map points from the image plane to a top-down representation of the bowling lane. Unlike a generic projective transformation, this mapping is constrained by the known physical dimensions of a standard bowling lane, namely a width of 1.066 m and a length of 19.16 m [27]. The destination coordinate system is defined directly in metric units, such that the rectified plane corresponds to a physically scaled representation of the lane surface. This establishes a direct correspondence between image coordinates and real-world measurements. The homography H is estimated such that:

$$\mathbf{x}_{\mathrm{metric}} = H\,\mathbf{x}_{\mathrm{image}}$$

where $\mathbf{x}_{\mathrm{metric}}$ lies in a metric coordinate system expressed in meters. As a result, the rectified output represents the lane in a perspective-corrected and metrically scaled coordinate system.

3.3.2 3D Trajectory Reconstruction

The reconstructed trajectory is obtained through a sequence of geometric transformations that map image-space detections into a 3D representation expressed in the coordinate system of the bowling lane.

First, the ball center is detected in the image plane as a sequence of positions over time, together with the corresponding estimated radii. The ball-lane contact point is then computed under the assumption that the ball remains in continuous contact with the lane surface, by correcting the detected center using the estimated radius along the vertical image axis.
These contact points are projected into the metric lane coordinate system using the previously computed homography, which is defined from the known physical dimensions of the lane. This yields a 2D trajectory expressed in real-world units on the lane plane.
Finally, a 3D representation of the motion is obtained by lifting the contact trajectory along the vertical axis by a distance equal to the ball radius, resulting in a set of 3D coordinates expressed in the lane reference frame. The final output of this stage is the trajectory of the ball expressed in the 3D coordinate system of the lane. The resulting representation is consistent with both the geometric constraints of the lane and the physical dimensions of the ball [6].

4. Experimental Activity and Analysis of Experimental Results

4.1. Preprocessing and Edge Detection Comparative Analysis

To empirically validate the optimal preprocessing and feature extraction strategy for the lane detection pipeline, a comparative analysis was conducted across nine distinct bowling video clips. These clips were specifically selected to encompass a variety of challenging real-world conditions, including varying camera exposures, surface glare, and heterogeneous lighting. The study evaluated the combinatorial performance of five single-channel transformations (Default Grayscale, Lightness, Hue, $R + G - B$, and PCA) paired with three standard edge detection algorithms (Sobel, Canny, and Laplacian). A trial was deemed successful strictly if the resulting gradient map enabled the subsequent Hough Transform and heuristic filters to correctly segment the entire playing surface by accurately isolating all four geometric boundaries (the foul line, the pin deck, and both lateral gutters), as demonstrated in Figure 6. The quantitative success rates for each combination are detailed in Table 1.

Figure 6: Criteria for segmentation success. Left: A successful trial where the estimated quadrilateral accurately isolates the true playing surface. Right: An unsuccessful trial where the left boundary is not correctly detected.

An analysis of the aggregated results reveals clear performance disparities not only among the color transformations but, crucially, among the edge detection algorithms. The Sobel operator emerged as the generally strongest edge detection method for correctly identifying lane borders, outperforming Canny and Laplacian across the most effective grayscale transformations. While the multi-stage Canny detector and the second-order Laplacian filter are exceptionally strong edge detectors, this extreme sensitivity proved detrimental in a visually cluttered bowling alley. Qualitative analysis of the failed clips revealed that Canny and Laplacian frequently detected crisp, high-contrast edges entirely outside the target playing surface, such as architectural lines on the background wall, adjacent lane gutters, or the bowler's silhouette. These highly defined but irrelevant structures generated overwhelmingly strong candidate lines in the Hough space. Consequently, these false positives bypassed the spatial and geometric heuristics of the pipeline, leading to incorrect boundary selection and severely deformed segmentation quadrilaterals (see Figure 7). Sobel, by providing a more subdued and directionally targeted gradient map, allowed the geometric filters to consistently lock onto the true lane boundaries.
When evaluating single-channel color transformations, PCA proved superior in both general robustness and peak performance. Unlike fixed-weight channel conversions, PCA dynamically projects color channels onto their first principal component to maximize the statistical variance of each image. This data-driven approach adapts

Table 1: Quantitative success rates of boundary detection for each preprocessing and edge detection combination across 9 video clips.

Grayscale Method	Edge Method	clip1	clip2	clip3	clip5	clip6	clip7	clip8	clip11	clip13	Sum
default	sobel	1	1	1	0	1	1	0	1	0	6
default	canny	0	0	0	0	0	1	0	1	0	2
default	laplacian	0	0	1	0	1	1	0	1	1	5
lightness	sobel	1	1	0	0	0	1	0	1	1	5
lightness	canny	1	0	1	0	0	0	0	1	0	3
lightness	laplacian	0	0	1	0	0	0	0	1	1	3
hue	sobel	0	0	0	0	0	0	0	0	0	0
hue	canny	0	0	0	0	0	0	0	0	0	0
hue	laplacian	0	1	1	0	0	0	0	0	0	2
r_g_minus_b	sobel	1	0	0	1	1	1	0	0	0	4
r_g_minus_b	canny	1	1	1	1	0	1	0	0	0	5
r_g_minus_b	laplacian	1	1	1	0	0	0	0	0	0	3
pca	sobel	1	1	1	1	1	1	0	1	0	7
pca	canny	1	0	0	1	1	0	0	1	0	4
pca	laplacian	0	0	1	0	1	1	0	1	1	5

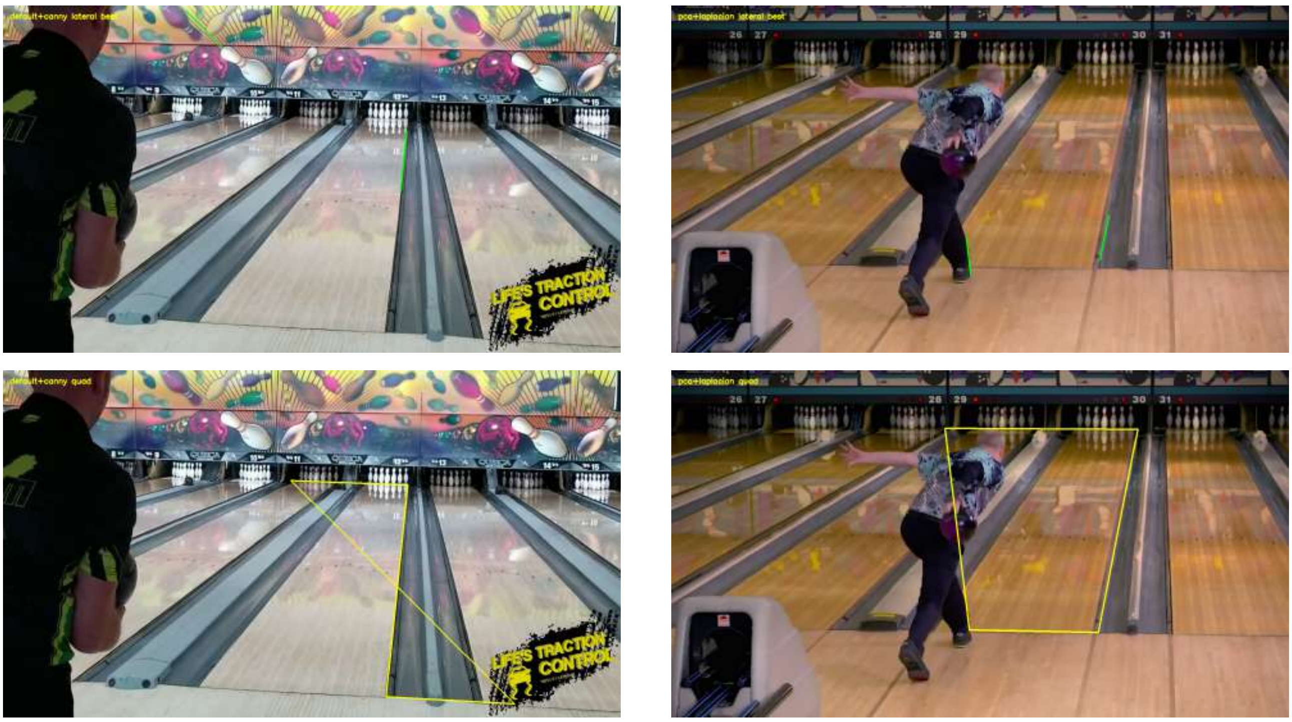

Figure 7: Examples of heuristic failure due to over-sensitive edge detection. Top Left: Canny detects background wall structures. Top Right: Laplacian detects the bowler's leg and adjacent lane. Bottom Row: The respective incorrect homography quadrilaterals resulting from these flawed edge maps.

effectively to the highly variable wood types, oil patterns, and lighting found in bowling alleys. By isolating dominant structural features (the lane boards) while suppressing localized lighting artifacts, PCA achieved the highest stability across all edge detectors. When paired with the directionally targeted Sobel operator, the pipeline reached its peak performance, successfully extracting accurate lane boundaries in 7 out of 9 video clips. Consequently, the PCA and Sobel combination was selected for the final boundary detection module.
Finally, the analysis provided valuable insights into the theoretical limitations of chromatic edge detection. It was initially hypothesized that the Hue channel from the HSV color space might yield interesting results by isolating pure color transitions and entirely negating the impact of absolute brightness and shadows. In practice, however, the Hue channel demonstrated the poorest performance of the study, completely failing to localize boundaries across almost all filters. Qualitative analysis of the output (see Figure 8) revealed the cause of this failure: the Hue representation effectively erased the critical geometric boundaries (particularly the bottom foul line) from the image entirely. This occurs because the physical demarcations of a bowling lane (such as a dark foul line or shaded gutters contrasting against the light wooden surface) are primarily defined by sharp shifts in luminance and intensity, rather than distinct shifts in core color. By stripping away the intensity information, the Hue channel eliminates the necessary gradient contrast, causing the borders to blend indistinguishably into the background. This visual evidence conclusively confirms that robust boundary extraction in a bowling setting relies fundamentally on structural intensity rather than pure chromaticity.

Figure 8: Output of the Hue channel on the lower lane segment. By discarding luminance data, the critical intensity contrast of the foul line is completely erased, resulting in a failure to detect the bottom boundary.

4.2. Top Boundary Localization via Object-Centric Regression

As established in the methodology, the top boundary of the lane (the pin deck interface) presents a unique geometric challenge: the physical edge is heavily occluded by the bowling pins themselves, making standard gradient-based edge detection ineffective. To overcome this occlusion, feature-based object detection methods, specifically SIFT, were initially evaluated for pin localization. While SIFT is generally highly robust to scale and rotation variations, it proved ineffective in this specific environment. The primary limitation arises from the small relative size of the pins within the global image frame at typical camera distances. Furthermore, because bowling pins possess a smooth, uniform surface with minimal internal texture, they generate an insufficient number of stable, high-contrast keypoints. Qualitative analysis of the feature matching outputs (see Figure 9) revealed that the algorithm consistently failed to localize the target pins on the deck. Instead, the limited keypoints extracted from the pin template produced spurious, incorrect matches with high-frequency noise and visually complex elements in the background, such as wall graphics. Due to this inability to reliably detect the target objects, feature-based extraction was discarded in favor of the intensity-based template matching approach, which successfully leverages the macro-structural silhouette of the pins rather than internal textural features.
For this reason, the pipeline employs multi-scale template matching to isolate the pins, extracting the bottom-center coordinates of their resulting bounding boxes as spatial candidate points for the boundary (see Figure 10).
The specific selection of the bottom-center coordinate, rather than the geometric centroid (midpoint) of the bounding box, is a critical design choice. Empirical analysis of the bounding box outputs revealed that using the centroid artificially elevated the regression line, mapping it across the midsections of the pins rather than the lane surface (Figure 11, Top Row). Because the physical objective is to map the pin deck interface, the bottom-center coordinate provides a geometrically accurate anchor directly to the true floor plane (Figure 11, Bottom Row).
While multi-scale template matching successfully localized the target pins, the raw output was inherently noisy. As anticipated in a visually complex bowling environment, the algorithm frequently generated false positive detections, predominantly from pin reflections on the highly polished synthetic oil and visually similar

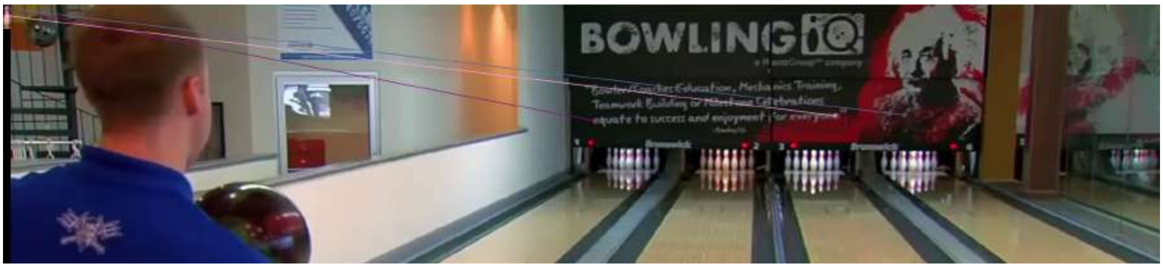

Figure 9: Failure of SIFT feature matching for pin detection. Due to the small scale and lack of internal texture on the pins, the algorithm fails to find valid corresponding keypoints on the pin deck. Instead, it generates spurious matches between the template (top left) and random high-frequency background elements.

Figure 10: Multi-scale template matching detections. The algorithm draws bounding boxes around identified bowling pins, capturing structural silhouettes on the target pin deck, adjacent lanes, and occasional background false positives.

Figure 11: Comparison of boundary candidate coordinate selection. Top Row: Utilizing the bounding box centroids (midpoints) artificially elevates the regression line, mapping it across the "necks" of the pins. Bottom Row: Utilizing the bottom-center coordinates correctly anchors the estimated regression line to the true floor plane at the pin deck interface.

architectural features in the background. Interestingly, the template matching also frequently detected pins actively standing in adjacent parallel lanes. Rather than treating these adjacent detections as noise, the pipeline exploits them to increase the stability of the boundary estimation. Because bowling alleys are constructed with coplanar, horizontally aligned pin decks, pins in adjacent lanes share the exact same geometric baseline as the target lane. Consequently, these adjacent detections serve as valid, extended data points that drastically improve the stability of the linear estimation.
To robustly extract the true boundary from this spatial data while filtering out the actual noise, a RANSAC regressor was evaluated. RANSAC is uniquely suited for this task due to its outlier rejection capabilities. By iteratively estimating a first-degree polynomial model from random subsets of the spatial coordinates, the algorithm successfully identified the dominant linear structure (the continuous base of the pins extending across the visible lanes) while ignoring coordinate data from reflections and background architecture that did not conform to this geometry.
Qualitative analysis of the regression output (see Figure 12) demonstrates the high efficacy of this approach. The RANSAC algorithm successfully segmented the raw candidate points into an inlier consensus set (visualized as blue points) and a rejected outlier set (visualized as red points). The inlier set actively incorporates valid pins from both the target and adjacent lanes, exploiting the extended horizontal baseline to correctly estimate the top boundary. The red spatial outliers encapsulate the false detections caused by reflections and background noise. By fitting the final line to these blue inliers, the pipeline generated an accurate and stable top boundary line (visualized in green). This confirms that decoupling the structural detection into an object-centric model effectively bypasses severe visual occlusions at the pin deck.

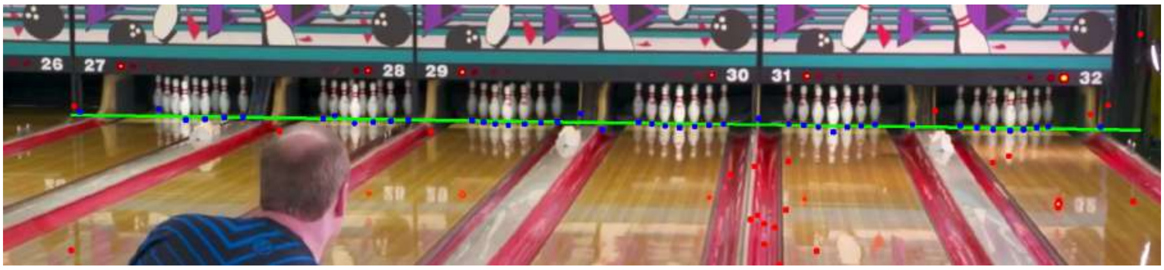

Figure 12: Performance of the RANSAC regressor on multi-scale template matching output. The bottom-center coordinates of all detected bounding boxes are plotted. The algorithm successfully classifies true pins (including those on adjacent, aligned lanes) as inliers (blue dots). It effectively rejects false positives from reflections and background structures as outliers (red dots). The final fitted polynomial (green line) accurately maps the occluded pin deck boundary using the extended geometric baseline.

4.3. Trajectory and Spin Detection Results

4.3.1 Spin and trajectory detection results

The ball detection pipeline was applied to eight clips across two release sessions. The releases of clips 9-12 and 13-16 were done consecutively with a non-homogeneous ball. For each clip, the lane polygon was reused for each session, across all clips within it. The *Circle Hough Transform* consistently detected the ball across all frames after preprocessing, with the weighted directed graph resolving ambiguous candidates into a coherent single trajectory. Cubic spline interpolation and Gaussian smoothing ($\sigma = 5$) filled the sparse detection gaps (as seen in Figure 13), and the final radius estimates followed the expected monotonic decrease as the ball travels away from the camera.

Figure 13: Final ball tracking visualization with and witouth occluded ball

The spin detection workflow was executed on all eight clips, yielding per-frame estimates of the rotation axis and angular velocity as shown in Table 2. The angular velocity θ is computed per consecutive frame pair via the Kabsch algorithm and averaged across the clip. The rev rate is derived as:

$$RPM = \frac{\theta \cdot fps}{2\pi} \cdot 60 \tag{14}$$

Table 2: Spin detection results across two recording sessions.

Session	Clip	Frames	FPS	Avg. θ (rad/frame)	Rev Rate (RPM)
A	clip_9	98	30	0.4420	128.1
	clip_10	125	30	0.4530	129.8
	clip_11	207	30	0.5075	145.4
	clip_12	109	30	0.4831	138.4
B	clip_13	196	30	0.3678	105.4
	clip_14	118	30	0.3637	104.2
	clip_15	104	30	0.3737	107.1
	clip_16	138	30	0.4063	116.4

The two sessions exhibit clearly distinct rev rate profiles. Session A yields a mean of *135.4 RPM* with a standard deviation of *7.8 RPM*, while Session B yields a mean of *108.3 RPM* with a standard deviation of *5.5 RPM*. The standard deviations are of 7.8 RPM and 5.5 RPM, 5.8% and 5.1%, respectively, indicating strong in-session consistency across clips of varying length. This repeatability suggests the pipeline is responding to a stable physical characteristic of each release rather than to measurement noise. The separation between sessions is consistent across all clips with no overlap: the lowest estimate in Session A (128.1 RPM, Clip 9) exceeds the highest estimate in Session B (116.4 RPM, Clip 16) by 11.7 RPM. This clean separation further supports the interpretation that the two sessions correspond to physically distinct release styles or bowlers, with Session A exhibiting a meaningfully higher rev rate.

Figure 14: Final visualization of ball's spin (yellow) and rotation axis (red).

4.4. Perspective Rectification and Homography Validation

Following the successful segmentation of the lane boundaries and the robust localization of the pin deck via RANSAC regression, the four intersecting corners of the playing surface (foul line and pin deck intersecting with the lateral gutters) are extracted to form a 2D quadrilateral in the image plane. To translate these distorted pixel coordinates into a uniform spatial environment, a planar homography matrix is computed. This projective transformation maps the source quadrilateral onto a normalized, top-down rectangular destination plane representing the physical proportions of the bowling lane.
To qualitatively validate the accuracy of the computed homography matrices, the original perspective video frames were warped to generate a direct bird's-eye view of the lane. As demonstrated in Figure 15, this transformation successfully rectifies the physical surface across various video clips characterized by differing camera positions, focal lengths, and lighting conditions. The geometric efficacy of the homography is visually confirmed by the structural alignment within these warped images: the longitudinal wood grain and target arrows maintain strict verticality without tapering toward a vanishing point. Because even minor pixel localization errors in the preceding boundary detection modules would severely distort this projection, the undistorted spatial structure achieved validates the stability of the foundational geometric model. Consequently, these homography matrices provide a reliable mathematical foundation for the subsequent mapping of the balls' real-world trajectories.

Figure 15: Bird's-eye view perspective rectification. The computed homography matrices eliminate perspective distortion, rendering the lane boards parallel. The horizontal alignment of the arrows and pins across all trials confirms that true metric proportions are successfully preserved.

4.5. Trajectory Reconstruction and 3D Spatial Visualization

Building upon the validated homography matrix and the continuously interpolated 2D tracking data, the final objective of the pipeline is to reconstruct the bowling ball's trajectory in real-world dimensions. As shown in

Figure 16, the interpolated trajectory is first visually rendered as a continuous trace directly onto the perspective video feed. Simultaneously, these continuous pixel coordinates are multiplied by the homography matrix to project the ball's positional path onto a two-dimensional bird's-eye view. It is important to note that, for graphical visualization purposes, the rendered top-down lane shown on the right side of the figure is intentionally distorted, compressed longitudinally and expanded laterally compared to true physical proportions. While the underlying homographic projection calculates metrically accurate spatial coordinates, mapping the projected trajectory onto this visually scaled lane significantly improves readability, exaggerating the transverse dynamics to make the ball's lateral hook clearly perceptible to the viewer.

Figure 16: 2D Trajectory mapping. Left: The interpolated tracking data is traced in red on the original perspective image plane. Right: The pixel coordinates are warped via the computed homography matrix into a normalized bird's-eye view. The graphical representation of the lane is intentionally scaled to exaggerate transverse movement, revealing the true lateral curvature of the throw.

To provide a comprehensive spatial analysis, this metrically accurate 2D planar data is subsequently extended into a full three-dimensional physical representation. The pipeline defines a real-world coordinate system anchored at the bottom-left corner of the physical lane (the origin), with the X and Y axes corresponding to the standardized physical length and width of the playing surface. The Z axis elevates the physical radius of the bowling ball above the floor plane, allowing for a volumetric representation of the movement.
Qualitative analysis of the final spatial visualizations (see Figure 17) confirms the success of the end-to-end pipeline. The reconstructed trajectory exhibits a smooth curvature that accurately mirrors the ball's hook dynamics from the original video. Ultimately, this proper spatial alignment demonstrates that the integrated computer vision pipeline successfully translates raw 2D pixel data into a reliable 3D kinematic model.

5. Conclusions and Future Work

As demonstrated in the experimental results, the proposed computer vision pipeline proved robust across its interdependent modules. The boundary detection module successfully localized the playing surface in highly varied environments, overcoming significant differences in wood textures, surface reflections, camera angles, and ambient illumination. Furthermore, the ball tracking architecture reliably isolated and tracked the ball's kinematic trajectory, utilizing temporal interpolation and graph-based filtering to maintain continuity even during periods of fast motion or occlusion by the bowler. The optical flow-based spin estimation module effectively captured the rotational dynamics, yielding consistent rev rate profiles that successfully differentiated the unique bowling capacities and release styles of two distinct players. Ultimately, the integration of planar homography and geometric priors allowed for a metrically accurate and visually coherent 3D reconstruction of the ball's trajectory, successfully translating raw 2D pixel data into a reliable spatial model.
However, while the proposed lane detection pipeline demonstrates robustness across various illumination conditions, wood textures, and moderate camera angles, the reliance on rigid geometric priors inherently limits its flexibility. Empirical testing (e.g., Clip 8) revealed that severe deviations in camera composition, perspective, and optical zoom cause the foundational edge detection heuristics to fail. To address these limitations in highly unconstrained environments, future work could explore the integration of Deep Learning architectures. Devel-

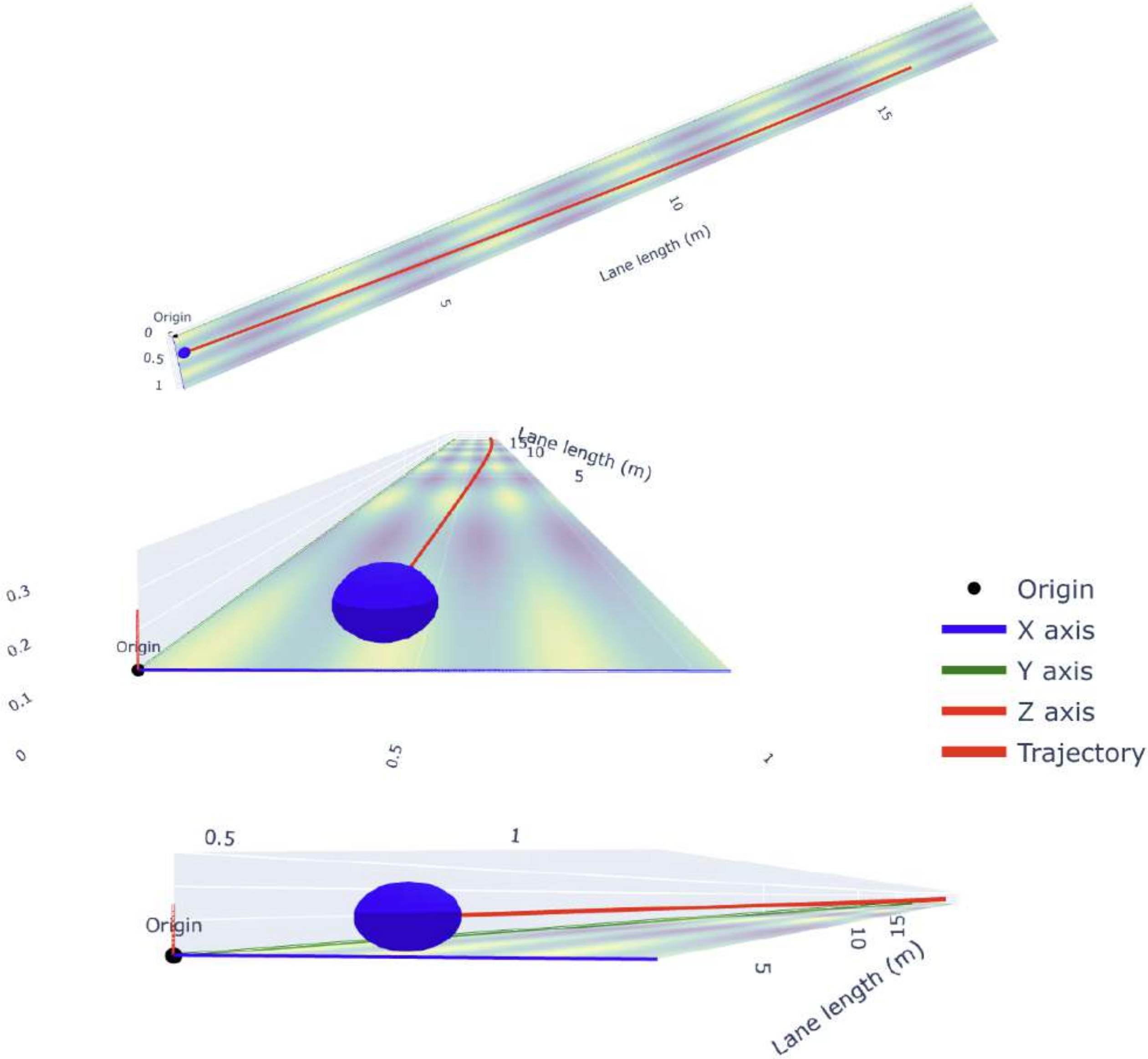

Figure 17: Three-dimensional spatial reconstruction of the lane and ball trajectory, shown from three distinct viewing perspectives to better illustrate the 3D spatial mapping.

oping a hybrid approach that combines the robustness of convolutional neural networks for initial boundary segmentation with our geometric algorithms for precise structural refinement presents a promising avenue for subsequent research.

Another limitation of the current 3D reconstruction stems from the planar assumption of the homography mapping, which mathematically restricts all tracked points to the floor plane ($Z = 0$). Consequently, if a bowler lofts the ball, the airborne pixel coordinates would be erroneously projected further down the lane, generating artificial spikes in the calculated spatial trajectory and velocity. To resolve this limitation, future research could evaluate the integration of physics-based motion priors to accurately estimate the ball's vertical height prior to surface contact.

Lastly, the system is constrained by the assumption of a static camera feed. Any camera movement would break the lane detection, trajectory detection, 3d reconstruction and spin detection modules, due to their underlying static hypothesis. To adapt the system for dynamic recording environments, subsequent research could explore frame-by-frame homography estimation or camera tracking algorithms to decouple sensor movements from the ball's actual trajectory and spin metrics

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